

Problem Sheet: Trig Substitutions 1

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Exercise 1. Using a trigonometric substitution, find the indefinite integral

$$\int \sqrt{1 - 4x^2} \, dx$$

Then state the interval of x values where this substitution is valid.

Exercise 2. Using a trigonometric substitution, compute

$$\int_{\frac{1}{2}}^1 \frac{1}{x\sqrt{1-x^2}} \, dx.$$

Exercise 3. Compute the following indefinite integral, assuming that $4 - x^2 \geq 0$ and $x \neq 0$.

$$\int \frac{1}{x^2\sqrt{4-x^2}} \, dx$$